

Started on Thursday, 5 September 2024, 2:57 PM**State** Finished**Completed on** Thursday, 5 September 2024, 3:21 PM**Time taken** 24 mins 17 secs**Grade** 100.00 out of 100.00

Question 1

Correct

Mark 20.00 out of 20.00

Create a class student with members name ,age,rollno and an user defined function show() to display the details of the student ,use the getter and setter method Information Hiding and conditional logic for setting an object attributes

For example:**Result**

Student Details: Jessa 10

Invalid roll no. Please set correct roll number

Student Details: Jessa 25

Answer: (penalty regime: 0 %)

Reset answer

```
1 class Student:
2     def __init__(self, name,__roll_no, age):
3         # private member
4         self.name = name
5         # private members to restrict access
6         # avoid direct data modification
7         self.__roll_no = __roll_no
8         self.__age = age
9
10    def show(self):
11        print('Student Details:', self.name, self.__roll_no)
12
13    # getter methods
14    def get_roll_no(self):
15        return self.__roll_no
16
17    # setter method to modify data member
18    # condition to allow data modification with rules
19    def set_roll_no(self, number):
20        if number > 50:
21            print('Invalid roll no. Please set correct roll number')
22        else:
```

	Expected	Got	
✓	Student Details: Jessa 10 Invalid roll no. Please set correct roll number Student Details: Jessa 25	Student Details: Jessa 10 Invalid roll no. Please set correct roll number Student Details: Jessa 25	✓

Passed all tests! ✓

Correct

Marks for this submission: 20.00/20.00.

Question 2

Correct

Mark 20.00 out of 20.00

Create two new, independent classes: **Turtle** and **Frog**. When you instantiate an object from the **Turtle** class, the object will use the **type** method as it is defined in that class. The same will be true of objects instantiated from the **Frog** class, despite the fact that the methods have the same name.

For example:**Result**

turtle
frog

Answer: (penalty regime: 0 %)

Reset answer

```
1 class Turtle:
2     def type(self):
3         print("turtle")
4
5 class Frog:
6     def type(self):
7         print("frog")
8         #define a function frog
9 obj_sea_turtle=Turtle()
10 obj_treefrog=Frog()
11
12 #Call the functions
13 obj_sea_turtle.type()
14 obj_treefrog.type()
```

	Expected	Got	
✓	turtle frog	turtle frog	✓

Passed all tests! ✓

Correct

Marks for this submission: 20.00/20.00.

Question **3**

Correct

Mark 20.00 out of 20.00

Write a Python program for simply using the overloading operator for adding two objects.

class name : product

For example:

Input	Result
80	amount : 179
99	location: chennaibanglore
chennai	
banglore	

Answer: (penalty regime: 0 %)

```

1 class product:
2     def __init__(self,x):
3         self.x=x
4     def __add__(self,u):
5         return self.x+u.x
6
7 obj1 = int(input())
8 obj2 = int(input())
9 amount = (f"amount : {obj1+obj2}")
10 str1 =str(input())
11 str2 =str(input())
12 location = (f"location: {str1+str2}")
13
14 print (amount)
15 print (location)
16

```

	Input	Expected	Got	
✓	80	amount : 179	amount : 179	✓
	99	location: chennaibanglore	location: chennaibanglore	
	chennai			
	banglore			

Passed all tests! ✓

Correct

Marks for this submission: 20.00/20.00.

Question **4**

Correct

Mark 20.00 out of 20.00

Write a python program to print the string with single and double quotes..

Place ' single quote with backslash when using single quote as delimiter

Place " double quotes with backslash when using double quote as delimiter

For example:

Result

Place ' single quote with backslash when using single quote as delimiter

Place " double quotes with backslash when using double quote as delimiter

Answer: (penalty regime: 0 %)

```
1 print ('Place \' single quote with backslash when using single quote as delimiter')
2 print ("Place \" double quotes with backslash when using double quote as delimiter")
```

	Expected	Got	
✓	Place ' single quote with backslash when using single quote as delimiter Place " double quotes with backslash when using double quote as delimiter	Place ' single quote with backslash when using single quote as delimiter Place " double quotes with backslash when using double quote as delimiter	✓

Passed all tests! ✓

Correct

Marks for this submission: 20.00/20.00.

Question **5**

Correct

Mark 20.00 out of 20.00

Define the abstract base class named Polygon and also define the abstract method. This base class inherited by the various subclasses. Implement the abstract method in each subclass. Create the object of the subclasses and invoke the **sides()** method.

For example:

Result
Triangle has 3 sides I have 4 sides Pentagon has 5 sides Hexagon has 6 sides

Answer: (penalty regime: 0 %)

Reset answer

```

1  from abc import ABC
2
3  class Polygon(ABC):
4
5      # abstract method
6      def sides(self):
7          pass
8
9  class Triangle(Polygon):
10
11
12      def sides(self):
13          print("Triangle has 3 sides")
14
15  class Pentagon(Polygon):
16
17      def sides(self):
18          print("Pentagon has 5 sides")
19
20      #Add code here
21  class Hexagon(Polygon):
22

```

	Expected	Got	
✓	Triangle has 3 sides I have 4 sides Pentagon has 5 sides Hexagon has 6 sides	Triangle has 3 sides I have 4 sides Pentagon has 5 sides Hexagon has 6 sides	✓

Passed all tests! ✓

Correct

Marks for this submission: 20.00/20.00.